

# LLMs: the Fundamentals

- Transformer architecture
- **GPT** in ChatGPT

# Learning Goals

Review

**Guest Speaker**

Generative

Pre-trained

Transformer

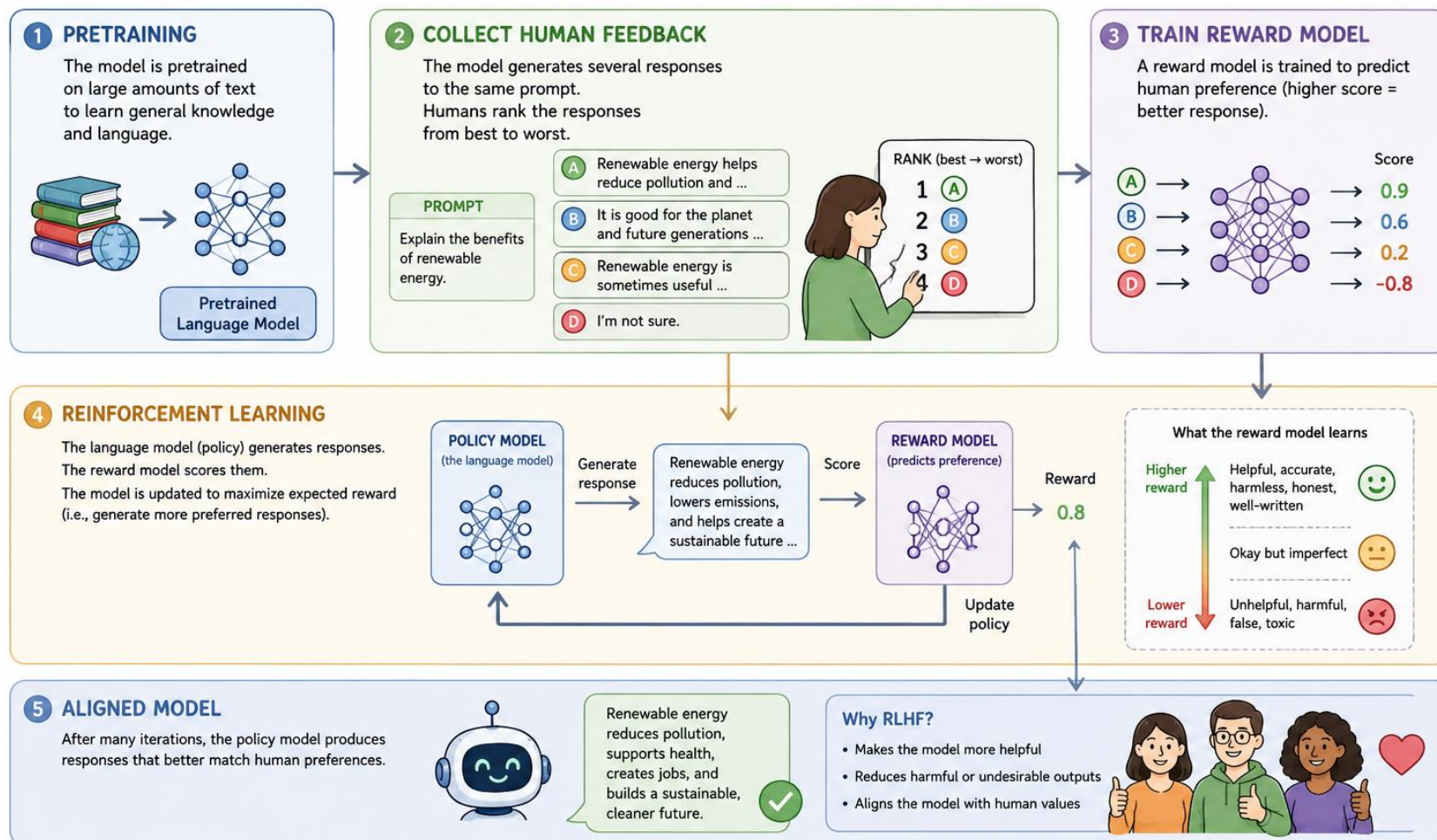
Human-in-the-loop



# The Alignment Problem: mitigated with Reinforcement learning from human feedback (RLHF)

## REINFORCEMENT LEARNING FROM HUMAN FEEDBACK (RLHF)

Aligning an AI model with human preferences



Note: In practice, RLHF also includes safety data, filtering, and iterative data collection.

[nature](#) > [news feature](#) > [article](#)

NEWS FEATURE | 02 July 2025

## AI ‘scientists’ joined these research teams: here’s what happened

Emerging ‘co-scientist’ systems use chatbots to mimic the deliberations of a research group. *Nature* asked researchers to test them out.

By [Nicola Jones](#)

[nature](#) > [articles](#) > [article](#)

Article | [Open access](#) | Published: 25 March 2026

## Towards end-to-end automation of AI research

[Chris Lu](#), [Cong Lu](#), [Robert Tjarko Lange](#), [Yutaro Yamada](#) , [Shengran Hu](#), [Jakob Foerster](#), [David Ha](#)  & [Jeff Clune](#) 

[Nature](#) **651**, 914–919 (2026) | [Cite this article](#)

275k Accesses | 26 Citations | 805 Altmetric | [Metrics](#)

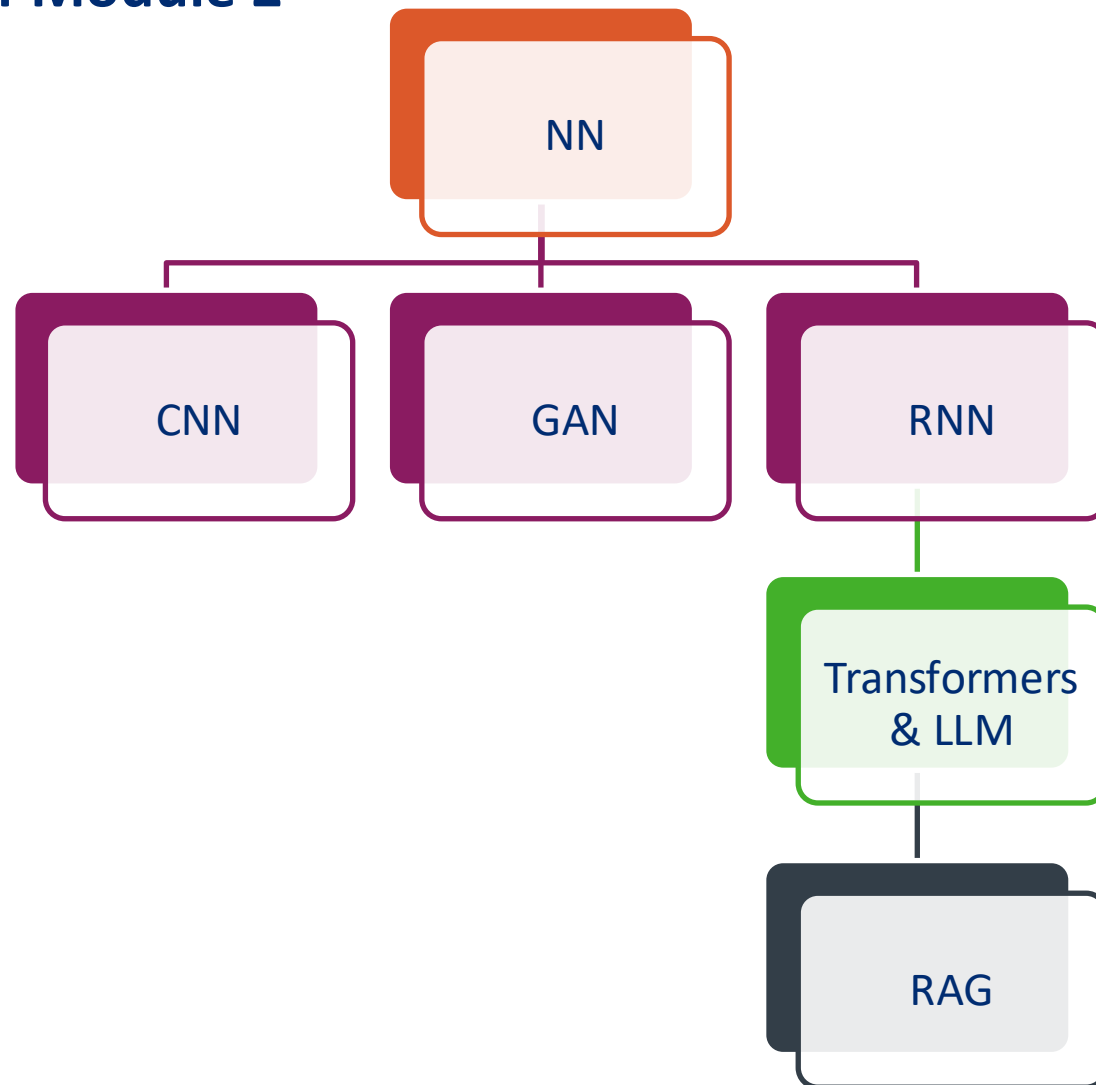
## The More You Automate, the Less You See: Hidden Pitfalls of AI Scientist Systems

Ziming Luo, Atoosa Kasirzadeh, Nihar B. Shah  
{zimingl, akasirza, nihars}@andrew.cmu.edu  
Carnegie Mellon University

## Learning Goals

- Review Transformer architecture.
- Define “Generative – Pre-trained – Transformer” (GPT) terminology
- Demonstrate tokenization & embeddings through live games (Semantis)
- Humans-in-the-loop

## ChatGPT - Review of Module 2



## Summary:

1. The goal of machine learning is to **learn patterns in data** and apply patterns **to predict** based on input
2. **Neural networks are a type of machine learning that allow arbitrarily complex relationships to scale**
  - A bunch of regressions stacked together
  - Billions of parameters

# (Review) Transformers

- **Self-Attention:** Global Pairwise Attention compared to CNNs (small local window)
  - Every token talks to every other token and considers every token pair  $(i, j)$
- Captures long-range information: **No locality bias**; Allows context
  - CNNs only look in neighborhood; RNNs only 'see' past context step by step

# Transformers

When do you need context? **JOKES**

An English professor, a physicist, and a statistician go out hunting. They spot a deer. The English professor writes a beautiful essay about the deer's elegance. The physicist carefully calculates the exact angle and muzzle velocity needed. The statistician says, "Let's assume the deer is in the crosshairs—then we'll see if we can hit it."

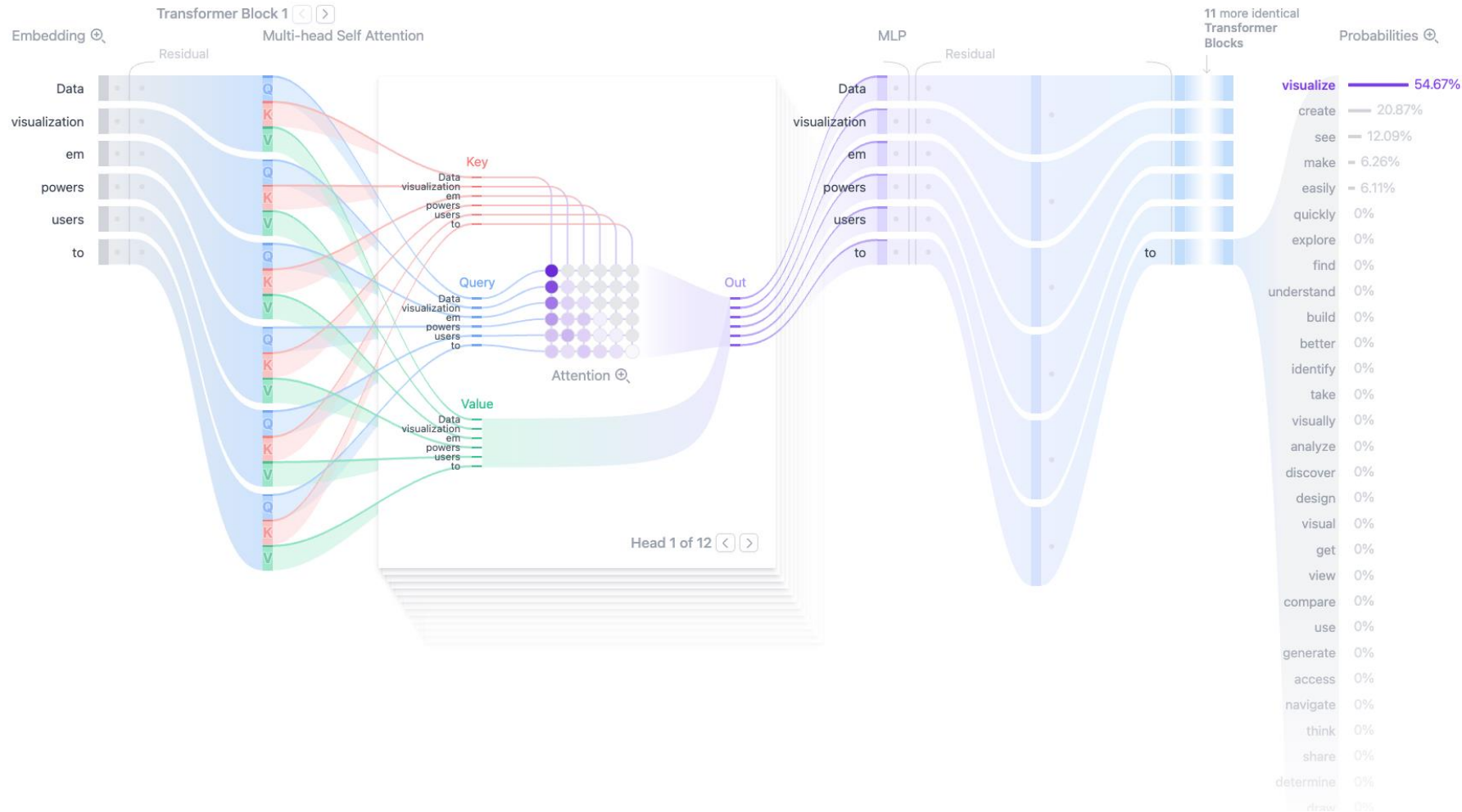
## Transformers

**Two hunters are out in the woods when one collapses...** He doesn't seem to be breathing, and his eyes are glazed. The other guy whips out his phone and calls the emergency services. He gasps, "I think my friend is dead! What can I do?" The operator says "Calm down. I can help. First, let's make sure he's dead." There is a silence, then a gun shot is heard. Back on the phone, the guy says "OK, now what?"

# How do transformer work in practice?

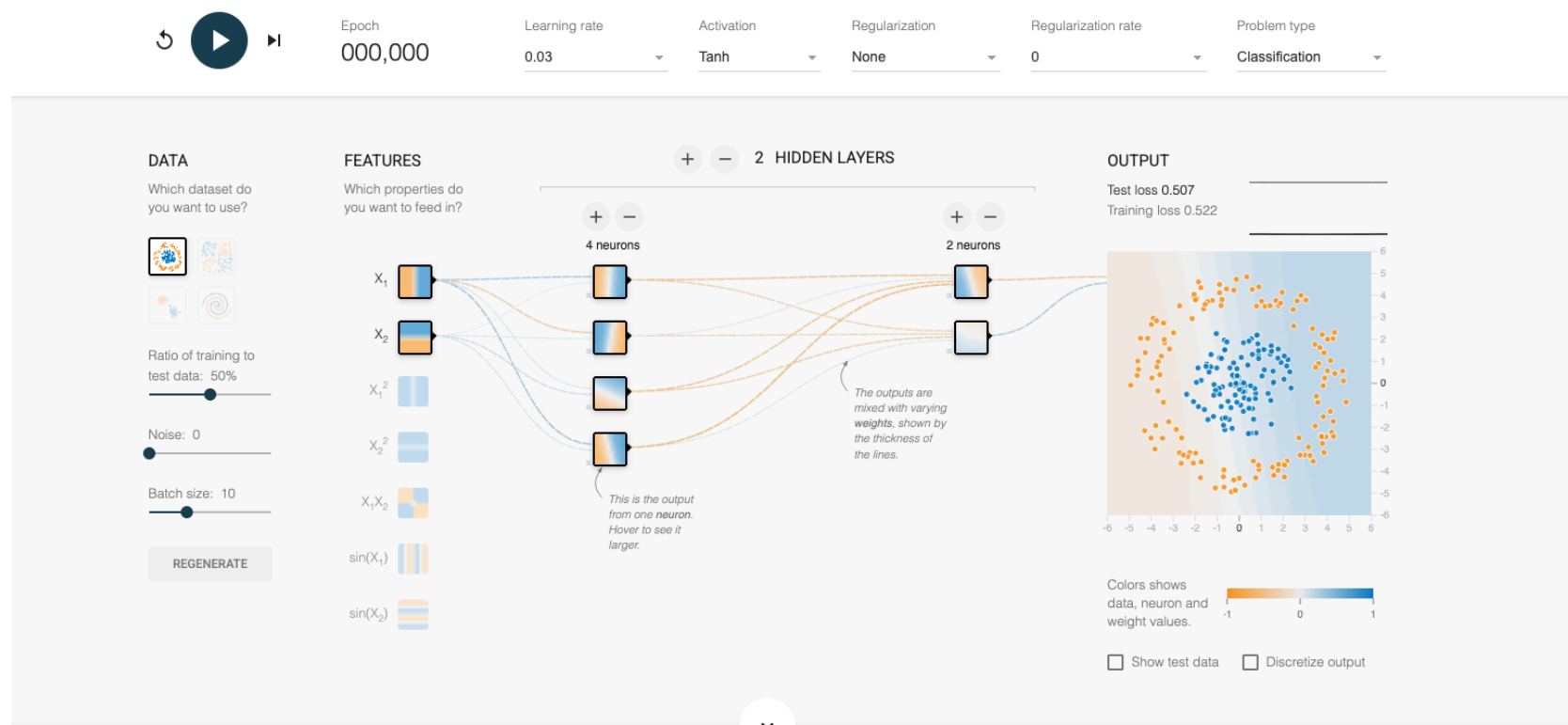
## TRANSFORMER EXPLAINER

Examples ▾ Data visualization empowers users to visualize Generate Temperature 0.8 Sampling ☒ Top-k ☐ Top-p k=5 PDF YouTube GitHub



# Simulating Neural Network parameters

Tinker With a **Neural Network** Right Here in Your Browser.  
Don't Worry, You Can't Break It. We Promise.



# What is a context window: What can LLMs 'remember'?

- A context window is the amount of information that an LLM can actively consider at one time when generating a response
- Think of it as **Working Memory**
  - More context = better reasoning, summation, conversations
- It is not permanent knowledge
  - Information may be dropped

| Human                | LLM               |
|----------------------|-------------------|
| Long-term Memory     | Pre-training data |
| Working Memory       | Context Window    |
| Looking something up | Retrieval (RAG)   |

# Transformers – Summary

- **Strengths:**

- Wide context
- Highly parallelizable
- Broadly applicable

- **Limitations:**

- Computationally expensive
- Memory footprint
- Requires training data

## To discuss *after* activity

We will be replaying the sentence game from yesterday but this time the words are ordered based on “importance” to the sentence

1. How did this compare to analyzing the sentence sequentially?
2. Does this way of ordering things introduce any new concerns?

# Sentence Analysis Activity

[https://docs.google.com/presentation/d/1FlefDxg3D\\_r3Exx\\_RX8d9H5tWXuLnIaiG1Mupc9Y2No/edit?slide=id.p#slide=id.p](https://docs.google.com/presentation/d/1FlefDxg3D_r3Exx_RX8d9H5tWXuLnIaiG1Mupc9Y2No/edit?slide=id.p#slide=id.p)

## To discuss *after* activity

We will be replaying the sentence game from yesterday but this time the words are ordered based on “importance” to the sentence

1. How did this compare to analyzing the sentence sequentially?
2. Does this way of ordering things introduce any new concerns?

## To discuss *after* activity

- How much context do you need to understand the meaning of a word in a sentence?
- Do you think there's a difference between how humans and machines understand language? If so, what is it?
- What might be the trade-offs between looking only at nearby information versus everything at once?
- In real-world AI systems (like translation or chatbots), why might it matter whether the model is using local or global attention?

# Match the Mindset

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Let's review!



- 
- A. Supervised learning** (e.g., labelled regression/classification)
  - B. Regularization techniques** (dropout, weight decay)
  - C. Over-fitting** (plus guardrails like validation sets)
  - D. Unsupervised learning** (e.g., K-means clustering)
  - E. Linear/logistic regression**
  - F. Self-attention / Transformer**
  - G. Generative Adversarial Network (GAN)**
  - H. Convolutional Neural Network (CNN)**
  - I. Recurrent Neural Network (RNN)**
  - J. “k-fold cross-validation”**
  - K. Evaluation metrics** (accuracy, precision-recall, etc.)
  - L. Activation function** (ReLU, sigmoid, etc.)

# BREAK

Come back at: XX:XX



# Guest Speaker: Research IT

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Ai at JAX



# BREAK

Be back by XX:XX



# ChatGPT



## LLMs

### ChatGPT

- Deep learning model
- **Generative:** predicts the next word
- **Pre-trained:** pre-trained for expectations based on massive amounts of text
- **Transformer:** type of neural network architecture (introduced in 2017, works by focusing attention onto relevant parts of the sentence)
  - We learned about transformers yesterday (and reviewed today)

How does a **L**arge **L**anguage **M**odel work?

**Generative:** predicts the next word

Q: Can we predict the missing words in the following sentence?

“Der schnelle braune fuchs springt \_\_\_\_\_”

How does a **L**arge **L**anguage **M**odel work?

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Q: Can we predict the missing words in the following sentence?

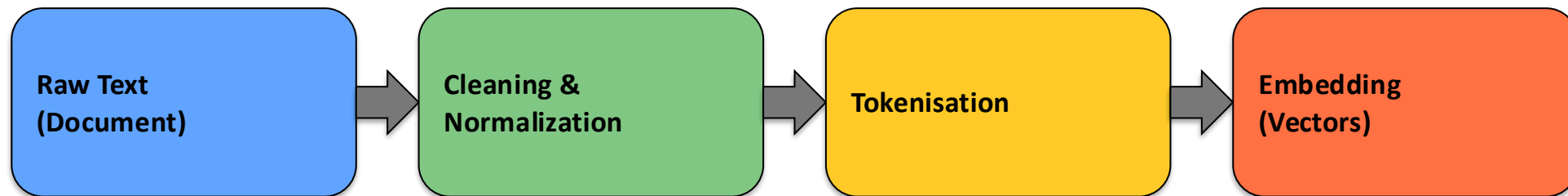
“The quick brown fox jumps \_\_\_\_\_”

How does a **Large Language Model** work?

**Generative:** predicts the next word

Q: Can we predict the missing words in the following sentence?

“Der schnelle braune fuchs springt \_\_\_\_\_”



## How does a **Large Language Model** work?

Q: Can we predict the missing words in the following sentence?

“The quick brown fox jumps \_\_\_\_\_”

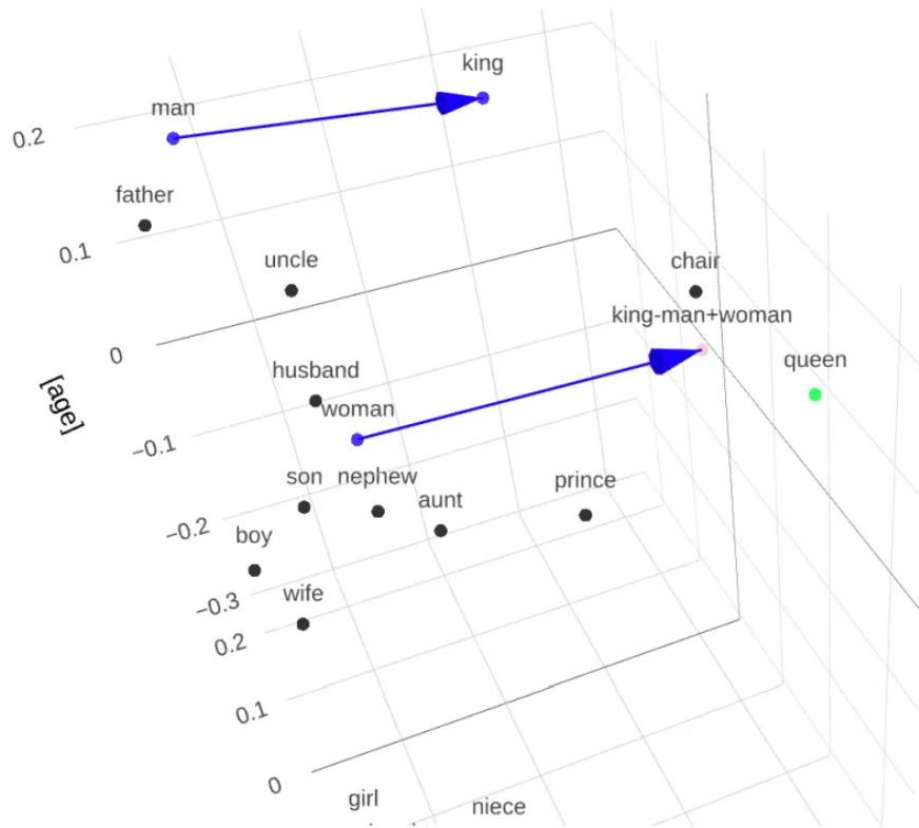
We are already familiar with the following:

Watch:

2-minute video on how Autocomplete works:  
[https://www.youtube.com/watch?v=us9tUY\\_yN7Y](https://www.youtube.com/watch?v=us9tUY_yN7Y)

- **Autocomplete:**
  - statistical n-gram (quantification of vector space of words)
  - 1–3-word context
  - Prediction target is the next word only

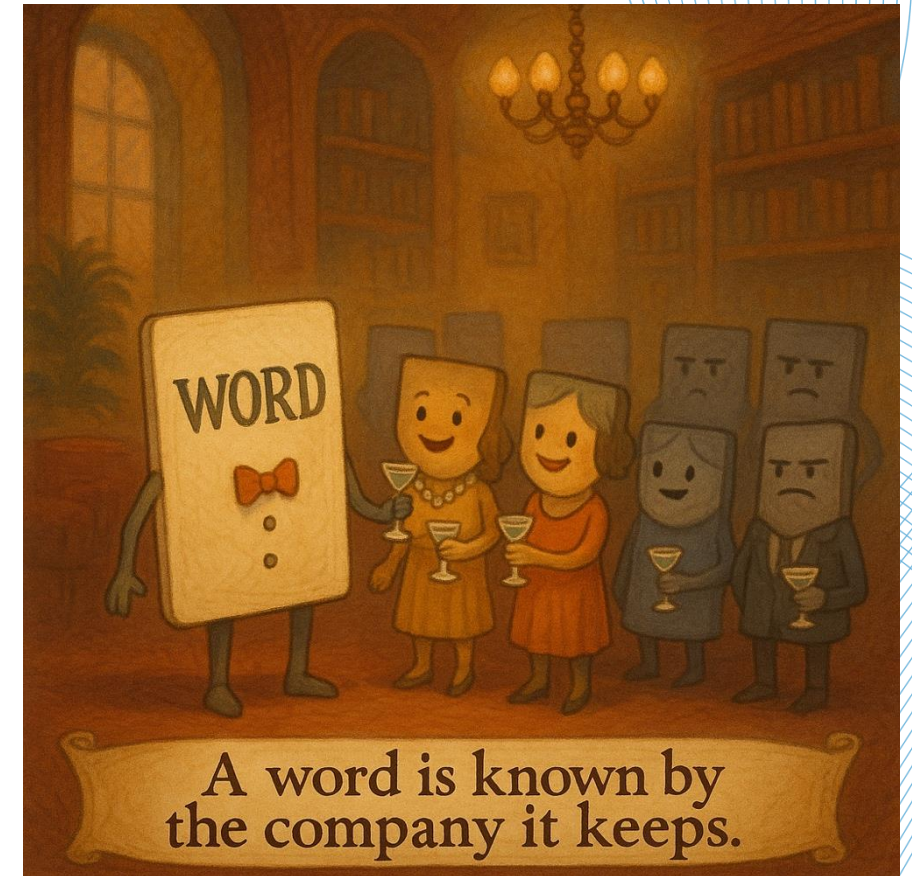
## Vector and Semantic Space



**“king – man + woman” = “queen”**

**Play:**

<https://research.google.com/semantris/>



## Vector and Semantic Space



How does a **Large Language Model** work?

**Generative:** mechanism is tokenization

Q: Can we predict the missing words in the following sentence?

“The quick brown fox jumps \_\_\_\_\_”

Play:

<https://research.google.com/semantris/>

More Challenging:

<https://semantle.com/>



## How does a Large Language Model work?

- Additional practice: <https://www.hooplaimpro.com/mind-meld>
- **N-grams:**
- Precursor to LMMs
  - **Trigram:**  $P(w_i | w_{i-2}, w_{i-1})$
  - Counts how often the three-word sequence,  $(w_{i-2}, w_{i-1}, w_i)$ , occurs
  - Captures dependencies within the three-word window
  - Problem: rare words
- Demo: <https://www.cs.cmu.edu/~dst/MarkovChainDemo/>

**“The best thing about AI is its ability to...”**

*The best thing about AI is its ability to*

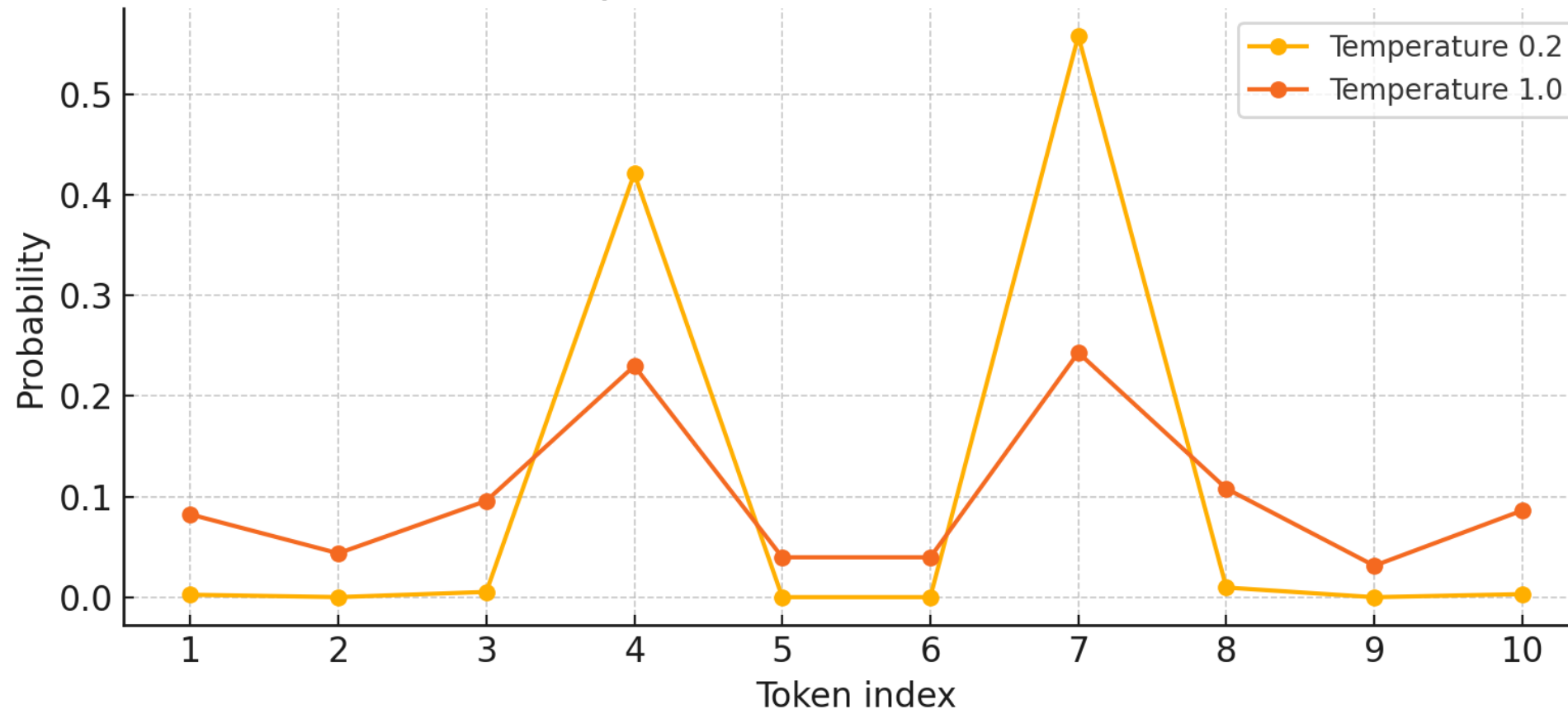
|            |      |
|------------|------|
| learn      | 4.5% |
| predict    | 3.5% |
| make       | 3.2% |
| understand | 3.1% |
| do         | 2.9% |

### Stochastic aspect:

- randomly doesn't pick the highest probability word ('temperature'=0.8)
- Same prompt can result in different results
- This is a ML classification task

“The best thing about AI is its ability to...”

Effect of Temperature on Next-Token Distribution



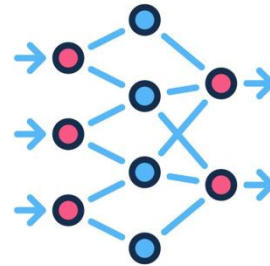
\* Many common models don't allow you to adjust this manually, just implicitly in your phrasing



## “The best thing about AI is its ability to...”

*The best thing about AI is its ability to*

|            |      |
|------------|------|
| learn      | 4.5% |
| predict    | 3.5% |
| make       | 3.2% |
| understand | 3.1% |
| do         | 2.9% |



{ The best thing about AI is its ability to,  
The best thing about AI is its ability to create,  
The best thing about AI is its ability to create worlds,  
The best thing about AI is its ability to create worlds that,  
The best thing about AI is its ability to create worlds that are,  
The best thing about AI is its ability to create worlds that are both,  
The best thing about AI is its ability to create worlds that are both exciting,  
The best thing about AI is its ability to create worlds that are both exciting, }

Because of the random choices, the same input prompt can result in different outputs:

The best thing about AI is its ability to learn. I've always liked the

The best thing about AI is its ability to really come into your world and just

The best thing about AI is its ability to examine human behavior and the way it

The best thing about AI is its ability to do a great job of teaching us

The best thing about AI is its ability to create real tasks, but you can

## ChatGPT

Let's see what ChatGPT will provide us as answers to the following questions:

1. The technical definition of a large language model (LLM) is:
2. Large language models (LLMs) can be used to create a variety of different outputs, such as:

*Let's compare these answers to the ones provided by Google Search autocomplete*

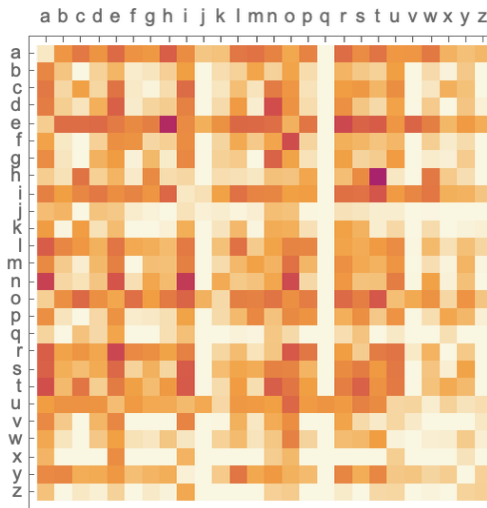
## LLMs

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100%

- ms) for
- r words,  
turn
-  39



2. Training on a large corpus of books, you can get probabilities for words, but you need a model that has parameters: billions of knobs to turn

## Pre-training in ChatGPT

- LLM are trained on huge amounts of data
- Allows them to predict next token
- Requires huge resources



VS



# Pre-training in ChatGPT

## Different types of Training

### 1. Pre-training (self-supervised): Knowledge

- Fed massive amount of information, but this only allows to babble (continuation), not **alignment**
- Not real 'learning' just regurgitating, like a fancy Google search autocomplete on steroids
- **Best Example: text summation**
- "Summarize this article in one sentence"

### 2. Instruction fine-tuning (supervised): Format

- models learn to respond, begin to **align**
- **High quality examples of instruction and response pairs**
- **Best Example: Answering questions**
- "In two sentences, compare Newtonian gravity and Einstein's general relativity"

### 3. Reinforcement learning (from human feedback): Judgement

- helps **alignment**
- **Critical for improving performance**
- **Human rate answers which is then used in subsequent iterations**

## LLMs

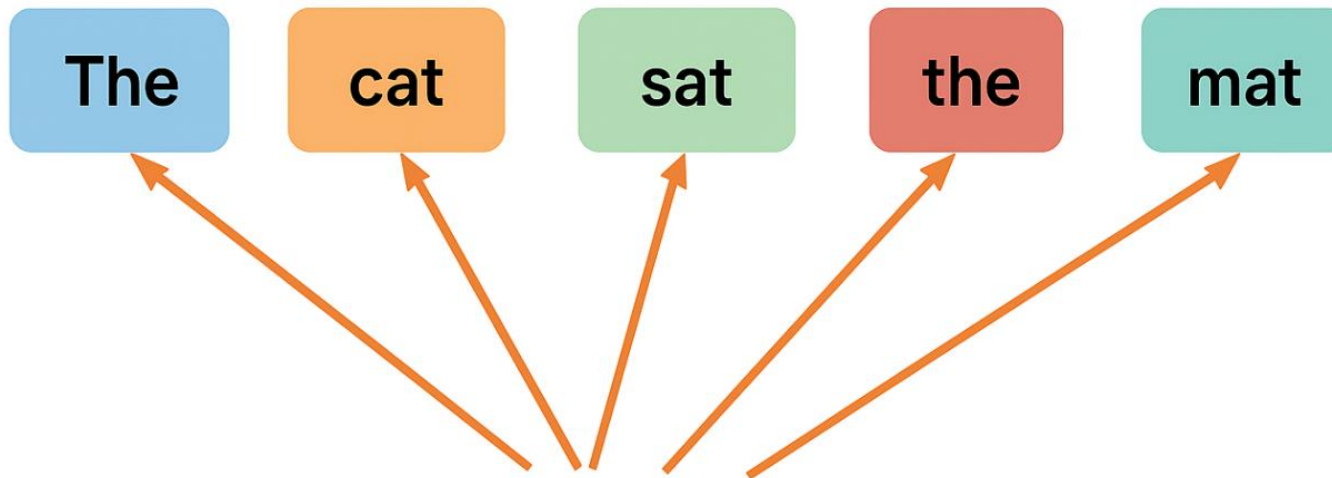
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## Transformers:

Self-attention: Global Pairwise attention

- every token talks to every other token and considers every token pair  $(i, j)$
- Captures long-range information: **No locality bias**; allows context



Self-attention: “cat” attends to every other word

# **Discernment: How do we evaluate the output of LLM?**

Human-in-the-loop

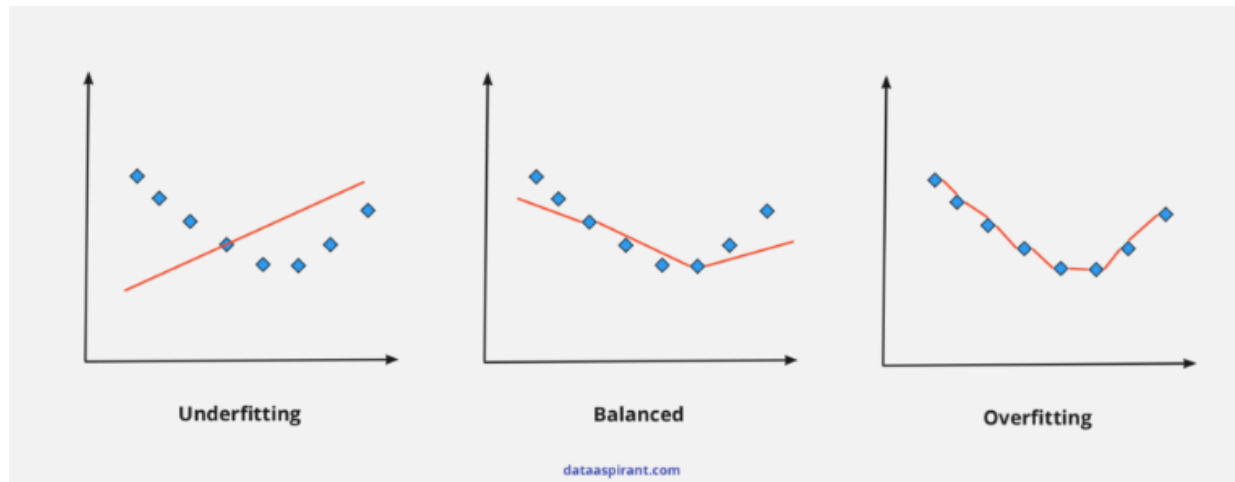


# Human-in-the-loop

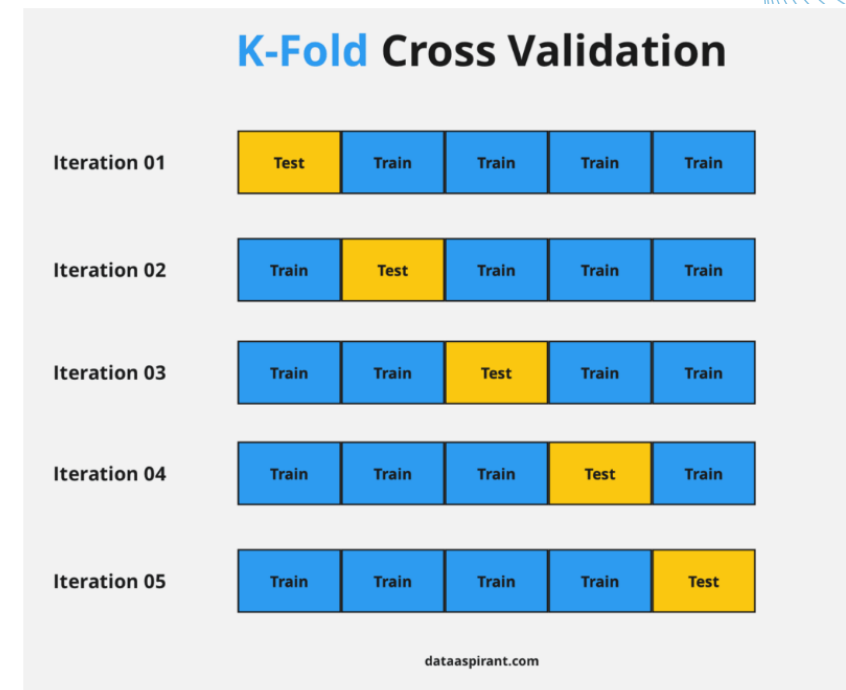
- **Training** versus **Testing**

## Strategy:

- split the data and use some of it for **training** the NN to **make predictions** and **reserve** some of it **to test** the NN



- K-fold cross-validation



## Discernment (& Evaluation)

- Domain expertise necessary (Human-in-the-loop) → can a human verify it?

## Alternative: Benchmark Tests

- Models answer thousands of standardized questions.
- Examples:
  - MMLU (Massive Multitask Language Understanding)
    - knowledge across many disciplines)
  - GPQA
    - graduate-level science
  - HumanEval (coding tasks)
  - MCAT, GRE, Bar exams

## Discernment (& Evaluation)

- Domain expertise necessary (Human-in-the-loop) → can a human verify it?
- General Safety Checklist:
  - **Factual accuracy**
    - Citation validity, facts being asserted
  - **Completeness & relevance**
    - Omitted alternatives
  - **Verifiable**
    - Does the logic 'make sense'?
  - **Uncertainty handling** (calibration)
    - Hidden assumptions
  - **Reproducibility**
    - Does the same prompt give similar answers?

## Expertise is CRUCIAL



# Human-in-the-loop

## Stroke: Vascular and Interventional Neurology



EDITORIAL ▶ Stroke Vasc Interv Neurol. 2026 Mar 3;6(2):e002321. doi: [10.1161/SVIN.126.002321](https://doi.org/10.1161/SVIN.126.002321)

### Jevons Paradox: Rise of the Machines May Create More Work for Physicians

[Jens Fiehler](#)<sup>1,8</sup>

▶ Author information ▶ Article notes ▶ Copyright and License information

PMCID: PMC12959442 PMID: [41815308](#)

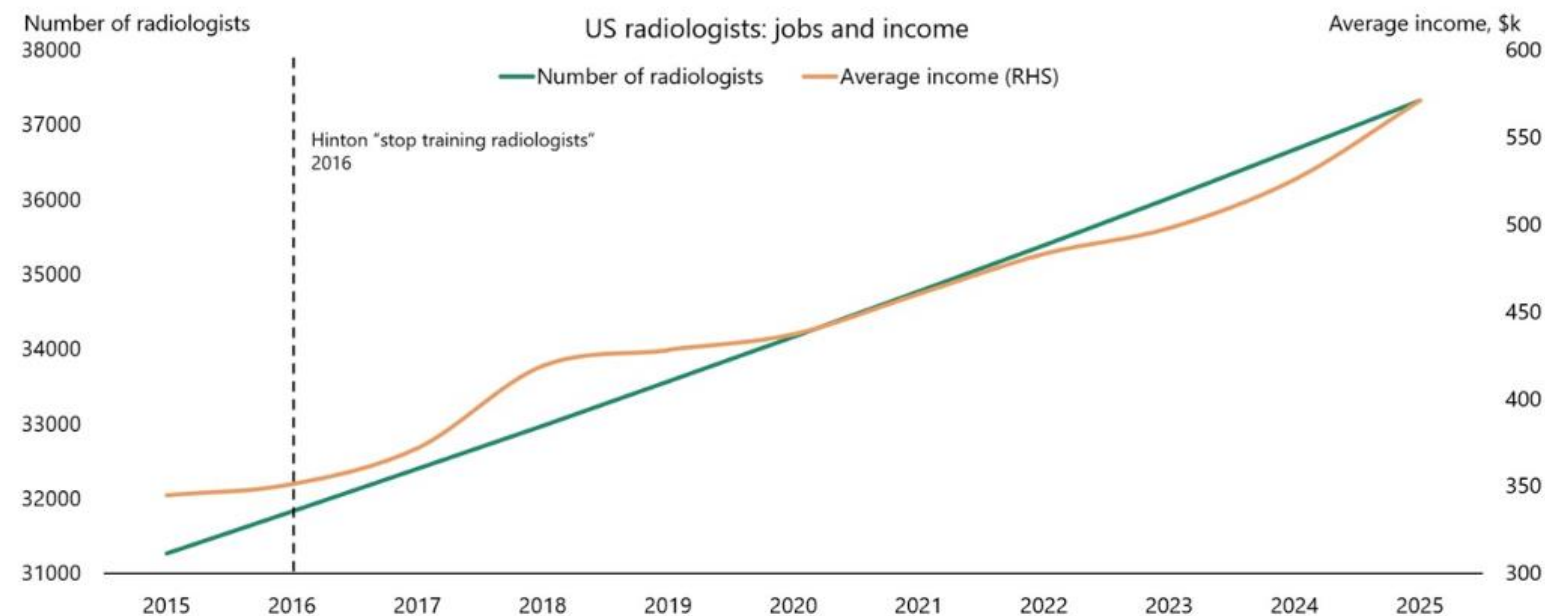
*It is a confusion of ideas to suppose that the economical use of fuel is equivalent to diminished consumption. The very contrary is the truth.*<sup>1</sup>

Stanley Jevons (1865)

*The surprising thing is the prediction that radiologists would be the first jobs to go was exactly the opposite. The trend shows that more radiologists are being hired now as a result of AI.*

Jensen Huang, CEO, Nvidia, November 19, 2025, U.S.-Saudi Investment Forum<sup>2</sup>

## The radiologist paradox

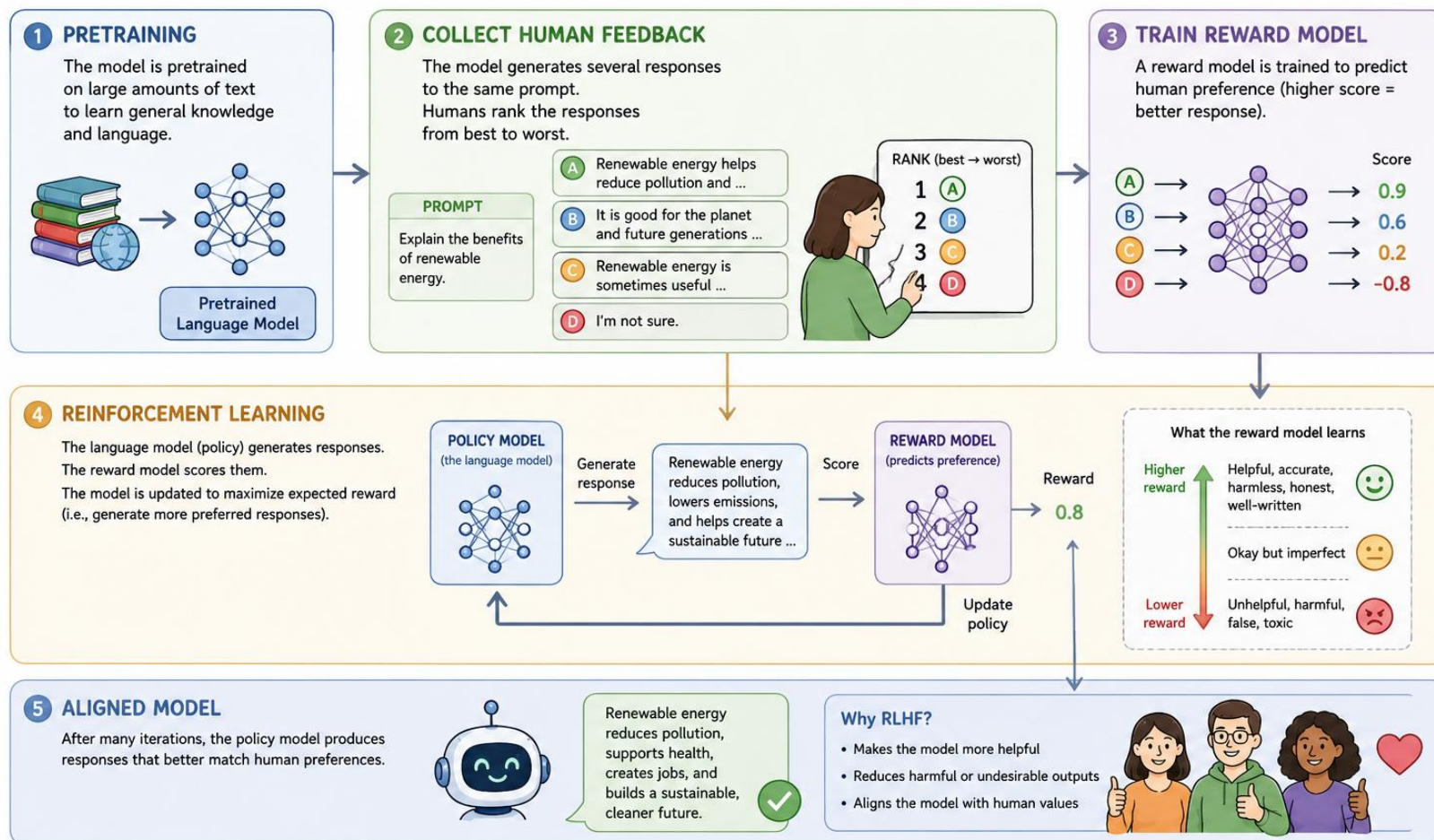


Sources: CMS Provider Data Catalog (National Downloadable Files), endpoints 30,723 (2014) and 36,024 (2023) per Rosenkrantz et al., AJR 2024; Medscape Physician Compensation Reports 2016–2026, Apollo Chief Economist

# The Alignment Problem: mitigated with Reinforcement learning from human feedback (RLHF)

## REINFORCEMENT LEARNING FROM HUMAN FEEDBACK (RLHF)

Aligning an AI model with human preferences



Note: In practice, RLHF also includes safety data, filtering, and iterative data collection.

## Trustworthiness

- Evidence from multiple measurements, and experts
- Not just one number

If ChatGPT answers 95% of medical questions correctly, should you trust it with *your* medical question?